

AWS Day 1

Topics covered:

- **Cloud vs. On-Premise:** Explored infrastructure differences, public vs. private network access (VPN/Private IPs), and multi-cloud providers (AWS, Azure, GCP).
- **AWS Global Infrastructure:** Covered Regions, Availability Zones (AZs) for high availability, and multi-account management using Landing Zones and Control Towers.
- **Core AWS Services:**
- **EC2:** Mastered pricing models (On-Demand, Reserved, Spot), Auto-Scaling limits, AMIs, instance types, and security configurations.
- **IAM & Security:** Clarified authentication vs. authorization, IAM policies (JSON structure), and ARNs.
- **VPC & S3:** Learned VPC components (Internet vs. NAT gateways, public/private subnets, route tables) and S3 storage classes.

On-Premise: physical on-premise servers

vs.

Cloud Computing using cloud-based infrastructure (3:14-4:34).

We will choose on premise or cloud depending on requirement, budget, and privacy standards.

On premise : Local not connected to the open internet

Cloud : Connected to the open internet

On-premises vs cloud

Both on premises and cloud both have hardware and are physically located somewhere.

Both are equally scalable.

Real difference?

Cloud server can be reached by an open internet Public IP

For connecting to On premises server we need a preferred IP (A VPM connection has to be established)

Also your IP will change, It will go into the **private IP range. 10. , 128. , 192. Class A B C**

- **Cloud Service Providers:**

There are different names for similar services in different clouds.

AWS vs AZURE vs GCP

Service Category	AWS	Azure	GCP	Utho	Oracle Cloud
Virtual Machines	EC2	Virtual Machines	Compute Engine	Cloud VM	Compute
Managed Kubernetes	EKS	AKS	Google Kubernetes Engine (GKE)	Managed Kubernetes	Kubernetes Engine (OKE)
Object Storage	S3	Blob Storage	Cloud Storage	Object Storage	Object Storage
Block Storage	EBS	Managed Disks	Persistent Disk	Block Storage	Block Volume
File Storage	EFS	Azure Files	Filestore	File Storage	File Storage
Managed SQL Database	RDS	Azure SQL Database	Cloud SQL	Managed Database	Autonomous Database / DB Systems
NoSQL Database	DynamoDB	Cosmos DB	Firestore / Bigtable	—	NoSQL Database
Data Warehouse	Redshift	Synapse Analytics	BigQuery	—	Autonomous Data Warehouse
Serverless Functions	Lambda	Azure Functions	Cloud Functions	—	Functions
Container Registry	ECR	Azure Container Registry	Artifact Registry	Container Registry	Container Registry
Load Balancer	Elastic Load Balancing	Azure Load Balancer	Cloud Load Balancing	Load Balancer	Load Balancer
Virtual Network	VPC	Virtual Network (VNet)	VPC	VPC / Private Network	Virtual Cloud Network (VCN)
DNS	Route 53	Azure DNS	Cloud DNS	DNS	DNS
CDN	CloudFront	Azure Front Door / CDN	Cloud CDN	CDN	Content Delivery Network
VPN	Site-to-Site VPN	VPN Gateway	Cloud VPN	VPN	VPN
Monitoring	CloudWatch	Azure Monitor	Cloud Monitoring	Monitoring	Monitoring
Identity & Access	IAM	Microsoft Entra ID / Azure RBAC	Cloud IAM	IAM	IAM
Secrets Management	Secrets Manager	Key Vault	Secret Manager	—	Vault
Key Management	KMS	Key Vault	Cloud KMS	—	Vault / KMS
Message Queue	SQS	Service Bus	Cloud Tasks / Pub/Sub	—	Queue
Pub/Sub / Messaging	SNS	Event Grid	Pub/Sub	—	Streaming
Email	SES	Communication Services	—	—	Email Delivery
Infrastructure as Code	CloudFormation	ARM / Bicep	Deployment Manager / Terraform	Terraform	Resource Manager / Terraform
AI/ML Platform	SageMaker	Azure Machine Learning	Vertex AI	—	OCI AI Services
API Management	API Gateway	API Management	API Gateway	API Gateway	API Gateway
Container Service	ECS	Container Instances	Cloud Run	Container Service	Container Instances
Data Streaming	Kinesis	Event Hubs	Pub/Sub	—	Streaming
Backup	AWS Backup	Azure Backup	Backup and DR	Backup	Backup

How do we determine which cloud to choose:

Quick mapping: AWS is generally the most extensive, Azure is particularly strong for Microsoft/enterprise environments, GCP is strong in data/AI and Kubernetes, Utho focuses on simpler cost-effective Indian cloud infrastructure, and Oracle Cloud is especially strong around Oracle databases and enterprise workloads.

Always check region and account ID before any action/changes

AWS Infrastructure:

Concept of *Regions* and *Availability Zones*

And their importance for high availability and disaster recovery (45:13-47:26).

Region > AZ (availability zone)

Region: Mumbai AP South-1

AZ: Az1, Az2, Az3

(The actual data sits here. Also we have the option to copy (Keep) the data in multiple Az for high availability (safety backup))

AWS

Landing zone and control tower.

For managing multi-account environments

Usually different accounts are maintained for different services for redundancy (ensure high availability)

Control Tower > Landing Zone

Point	Landing Zone	AWS Control Tower
Definition	A multi-account AWS foundation/architecture	An AWS managed service for creating and governing a Landing Zone
Main Purpose	Provides a secure, standardized AWS environment	Automates setup and ongoing governance
Multi-Account	✔ Yes	✔ Yes
AWS Organizations	Used as part of the architecture	Uses and manages AWS Organizations
Account Creation	Custom setup/automation	Account Factory
Governance	Mostly custom	Built-in Controls / Guardrails
Security	Designed and configured by the organization	Provides standardized governance mechanisms
Logging	Can be configured	Integrated centralized logging capabilities
Identity	Configure separately	Integrates with IAM Identity Center
Customization	✖ High	Medium
Automation	Depends on implementation	✖ High
Management Effort	Higher	Lower
Best For	Custom AWS architectures	Standardized enterprise AWS environments
Relationship	The foundation/architecture	The service used to establish and govern it

Detailed explanations of AWS services like EC2, IAM, and S3.

EC2 discussion with their pricing model, instance types, auto scaling, AMI.

EC2 Flexible pricing models:

On-demand, (expensive)

Reserved, (highly discounted, upfront payment for 1 year or 3 year)

Spot Instances, (used for RnD)

Auto Scaling for dynamic resource adjustment based on traffic.

Levels of auto-scaling:

Desired

Minimum limit

Maximum limit

AMI (Amazon Machine Image):

Pre configured virtual machines

Instance type:

(t3.micro etc)

hardware on which our virtual machine will actually run

Misc.

Key-pair : Is has to be created to be able to login to our instance

Auto-assign public IP: to be able to access our instance from internet

Security group: every application has a particular security group

IAM role

Difference between authentication and authorisation

concept of policies in AWS IAM.

Authentication vs Authorization

who you are vs what can you do

Policies < Roles

2:07:00

2:39:30

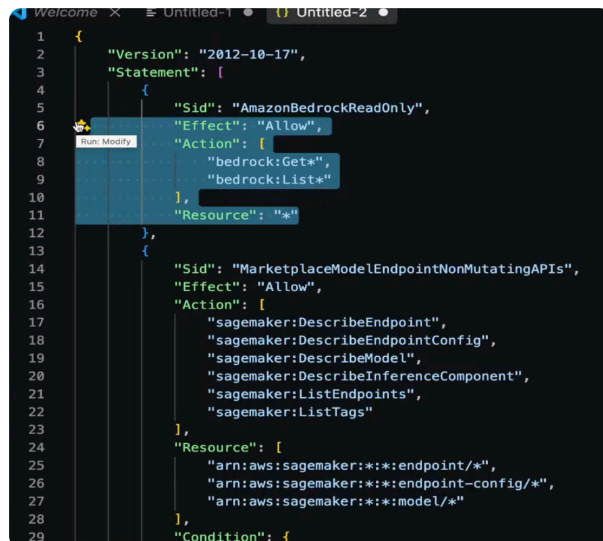
Actual power lies with the policy

"Effect"

"Action"

"Resource"

JSON file of a policy:-



```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "AmazonBedrockReadOnly",
6       "Effect": "Allow",
7       "Action": [
8         "bedrock:Get*",
9         "bedrock:List*"
10      ],
11      "Resource": "*"
12    },
13    {
14      "Sid": "MarketplaceModelEndpointNonMutatingAPIs",
15      "Effect": "Allow",
16      "Action": [
17        "sagemaker:DescribeEndpoint",
18        "sagemaker:DescribeEndpointConfig",
19        "sagemaker:DescribeModel",
20        "sagemaker:DescribeInferenceComponent",
21        "sagemaker:ListEndpoints",
22        "sagemaker:ListTags"
23      ],
24      "Resource": [
25        "arn:aws:sagemaker::*:endpoint/*",
26        "arn:aws:sagemaker::*:endpoint-config/*",
27        "arn:aws:sagemaker::*:model/*"
28      ],
29      "Condition": {
```

Concept of **Amazon Resource Name(ARN)** and how it is used to identify a resource in aws.

ARN: It is a unique identifier used to identify an AWS resource

Virtual Private Cloud(VPC)

Important interview question(on components of VPC):

NAT gateway v/s Internet gateway,

Aspect	NAT Gateway	Internet Gateway
Purpose	Lets private subnet resources reach the internet outbound	Lets public subnet resources communicate both ways with the internet
Direction	One-way, outbound only (private resources can't be reached from internet)	Two-way, inbound and outbound
Used by	Private subnets (DB servers, backend instances)	Public subnets (web servers, load balancers)
IP handling	Translates private IPs to a single NAT IP for outbound traffic	Maps public IP directly to the instance (1:1)
Placement	Sits in a public subnet, referenced by private subnet's route table	Attached directly to the VPC
Cost	Charged per hour + data processed	No additional charge (in most cloud providers like AWS)

Scalability	Managed, auto-scales (in AWS's managed NAT Gateway)	N/A, it's just a routing target, not a scaled resource
Security	Blocks unsolicited inbound connections by design	Doesn't filter traffic on its own, security groups/NACLs handle that
Example use case	App server in private subnet needs to download an OS patch	Web server in public subnet needs to be reachable by users

The core distinction:

NAT gateway is a one-way door for private resources going out,
Internet gateway is the actual front door for anything public-facing.

[Difference b/w public vs private subnet.](#)

Aspect	Public Subnet	Private Subnet
Internet access	Direct, via Internet Gateway	No direct access, only outbound via NAT Gateway (if configured)
Reachable from internet	Yes, resources with public IPs can be accessed directly	No, resources aren't reachable from outside
Typical use case	Web servers, load balancers, bastion hosts	Databases, internal app servers, backend systems
IP assignment	Public IP or Elastic IP assigned	Only private IP, no public IP
Route table	Routes 0.0.0.0/0 to Internet Gateway	Routes 0.0.0.0/0 to NAT Gateway (or no internet route at all)
Security exposure	Higher, directly exposed to internet traffic	Lower, shielded behind the public layer
Example resource	Public-facing website, API gateway	RDS database, internal microservices

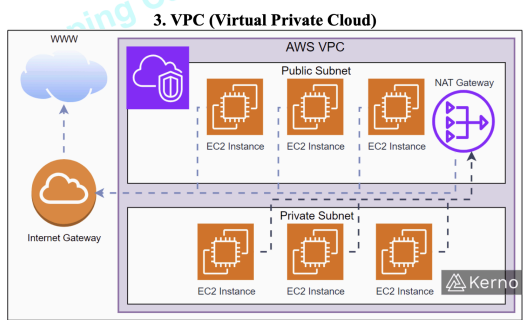
Basically:

public subnet is what the internet can see and touch directly,

private subnet is what stays hidden behind it and only reaches out when it needs to (patches, updates), ~~never the other way around.~~

Route table of Private subnet has route to NAT Gateway

Route table of Public subnet has route to Internet Gateway



- **Purpose:** Creates isolated networks in AWS for resources.

Migration vs Patching

CR: change request

Simple Storage Service(s3)

Interview ques: Storage classes in S3

Storage Class	Best For	Access Frequency	Minimum Storage Duration	Availability	Key Point
S3 Standard	Frequently accessed data	Frequent	None	99.99%	Default / general purpose
S3 Intelligent-Tiering	Data with changing access patterns	Unknown / changing	None	99.90%	Automatically moves data between tiers
S3 Standard-IA	Infrequently accessed data	Less frequent	30 days	99.90%	Lower storage cost, retrieval fee
S3 One Zone-IA	Infrequent, re-creatable data	Less frequent	30 days	99.50%	Stored in one AZ
S3 Glacier Instant Retrieval	Archive data needing quick access	Rare	90 days	99.90%	Retrieval in milliseconds
S3 Glacier Flexible Retrieval	Long-term archives	Rare	90 days	99.99%*	Retrieval from minutes to hours
S3 Glacier Deep Archive	Very long-term archives	Very rare	180 days	99.99%*	Lowest-cost S3 storage
S3 Express One Zone	High-performance workloads	Very frequent	None	99.95%	Very low latency, single AZ

▼ Misc. cloud

IAAS

Infrastructure - hardware

PAAS

Platform - hardware + OS

SAAS

Software - hardware + OS + software/applications

On premise - Not connected to the internet (being user internally)

Cloud - Connected to the network

On spot server

(used for testing purposes)

Terminates on a notice of 2 minutes

AWS - EC2

AZURE - Virtual machine

GCP - Google compute engine

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Not easy to build/scale

Vertical scaling = Same resource increase

(Downtime possible)

Easy to scale

Horizontal scaling = Increasing the number of instances

(No Downtime)

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Account ID & Region

(Always remember the region where you have created the instance)

US East-1 is the cheapest service

t3.Micro (free forever)

KeyValue pair??

- Key: Represents the category or the name of the attribute (e.g., Environment or Owner).
- Value: Represents the specific data assigned to that key (e.g., Production or DevOpsTeam).

Instance naming convention for instance:

<Company name> <Region> <Appplication> <Team
working upon> <Name>

S3 has region specific name now??

Account regional namespace (new)